

Let us emphasize that the issues arising in the design and analysis of numerical methods for molecular dynamics are slightly different than those confronted in other application areas. For one thing the systems involved are highly structured having conservation properties such as first integrals and Hamiltonian structure. We address the issues related to the inherent structure of the molecular N-body problem in both this and the next chapter; wherein we shall learn that *symplectic* discretizations are typically the most appropriate methods.

Another special aspect of the molecular system is that normally it is sensible to use a fixed stepsize, that is each timestep corresponds to a fixed interval of real time. This is in contrast to other applications where it is found to be important to vary the step during simulation. The reason that a uniform stepsize is used is because, in a simulation of many particles, the complexity of the system ensures that if a strong local force is encountered in some corner of the system, a force of similar magnitude will be found somewhere else at the next instant. Even if, occasionally, an instantaneous event is observed that could be controlled by reducing the stepsize, the necessary adaptive machinery can impair the geometric properties and reduce the efficiency of the numerical procedure.² There is no trivial way of selecting the stepsize a priori. In typical practice, one performs several trial runs, examining the fluctuations in energy or other easily computable quantities and makes the choice of stepsize in order to keep these within a tolerable range. The molecular dynamics timesteps are typically quite small, measured in femtoseconds, in order to capture the rapid fluctuations of the atoms (in Chap. 4, we discuss ways of increasing the timestep).

2.1 Error Growth in Numerical Integration of Differential Equations

In this section we discuss the issues of convergence and accuracy in numerical integration methods for solving ordinary differential equations.

Let us begin by considering Euler's method in a bit more detail to understand its convergence order. The *convergence* of a numerical method refers to the ability of the method to provide an arbitrary level of accuracy by using small enough timesteps. The *order of accuracy* is the exponent in the power law by which the error in the method is related to the stepsize. For example, when we say that a method is third order accurate, we mean that the global error (on a fixed finite time interval) can be bounded by Kh^3 , where h is a sufficiently small timestep and K is a number which depends on the length of the time interval and the features of the problem, but which is independent of h .

The flow map approximation in the case of Euler's method is

²A variable stepsize is only used where one expects extreme changes in particle velocities over the course of a simulation (see e.g. [75, 390] for examples arising in radiation damage cascades).

$$\mathcal{G}_h(\mathbf{z}) = \mathbf{z} + h\mathbf{f}(\mathbf{z}),$$

and we have

$$\mathbf{z}_{n+1} = \mathcal{G}_h(\mathbf{z}_n), \quad n = 0, 1, 2, \dots \quad (2.2)$$

with $\mathbf{z}_0 = \boldsymbol{\xi}$. The points $\{\mathbf{z}_n\}$ are intended to be approximations to the values of the solution. The obvious question is this: how good an approximation is $\mathbf{z}_n$ of $\mathbf{z}(nh)$?

Let the approximate solution vectors at successive timesteps be $\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_v$ where $vh = \tau$. We assume that τ , the length of the time interval, is fixed, and v is an integer parameter representing the total number of timesteps. In order to improve the quality of the approximation, the parameter v may be increased, as the stepsize is proportionately decreased. The error at step n is defined by $e_n = \|\mathbf{z}_n - \mathbf{z}(t_n)\|$, where $t_n = nh$; it clearly depends on h . With these definitions one can prove the following result, which is typical of the sorts of results that are available for numerical methods for initial value problems:

Theorem 2.1 *Let $\mathcal{D}$ be a bounded, open region in $\mathbb{R}^m$ such that $\mathbf{f} : \mathcal{D} \rightarrow \mathbb{R}^m$ is continuously differentiable. Let $\boldsymbol{\xi}$ be an interior point of $\mathcal{D}$ and suppose the initial value problem (2.1) has a unique solution that remains in $\mathcal{D}$ for $t \in [0, \tau]$. Then there exists a constant $C(\tau) > 0$ such that for sufficiently large $v \in \mathbb{N}$ the numerical solution $\mathbf{z}_n$ remains in $\mathcal{D}$ for $n = 0, 1, \dots, v$, where $hv = \tau$, and, moreover, the maximum global error in Euler's method satisfies*

$$\bar{e} := \max_{0 \leq n \leq v} e_n \leq C(\tau)h.$$

In short, the error in the approximation obtained on $[0, \tau]$ is reduced in direct proportion to the number of steps taken to cover this interval. Another way to say this is that $\bar{e} \propto h$, or, using the order notation, $\bar{e} = \mathcal{O}(h)$. Because the global error is of order h^r , where $r = 1$, we say that Euler's method is a *first order* method, or that it *converges with order $r = 1$* .

2.1.1 Example: The Trimer

Let us test Euler's method on the trimer model (formulated as a system in $\mathbb{R}^4$). Fixing initial values ($\mathbf{q}_0 = (0.5, 1.0); \mathbf{p}_0 = (0.1, 0.1)$) we solve the trimer on $[0, 4]$ using the Euler method first with 40 time-steps of length 0.1, next using 80 time-steps of length 0.05 and then with 160 steps of size 0.025, etc. Each calculation results in a different “discrete trajectory.” The solutions are graphed in the xy projection in the left panel of Fig. 2.2 with line segments connecting the points; these piecewise linear paths take on the appearance of smooth curves as the stepsize is reduced.